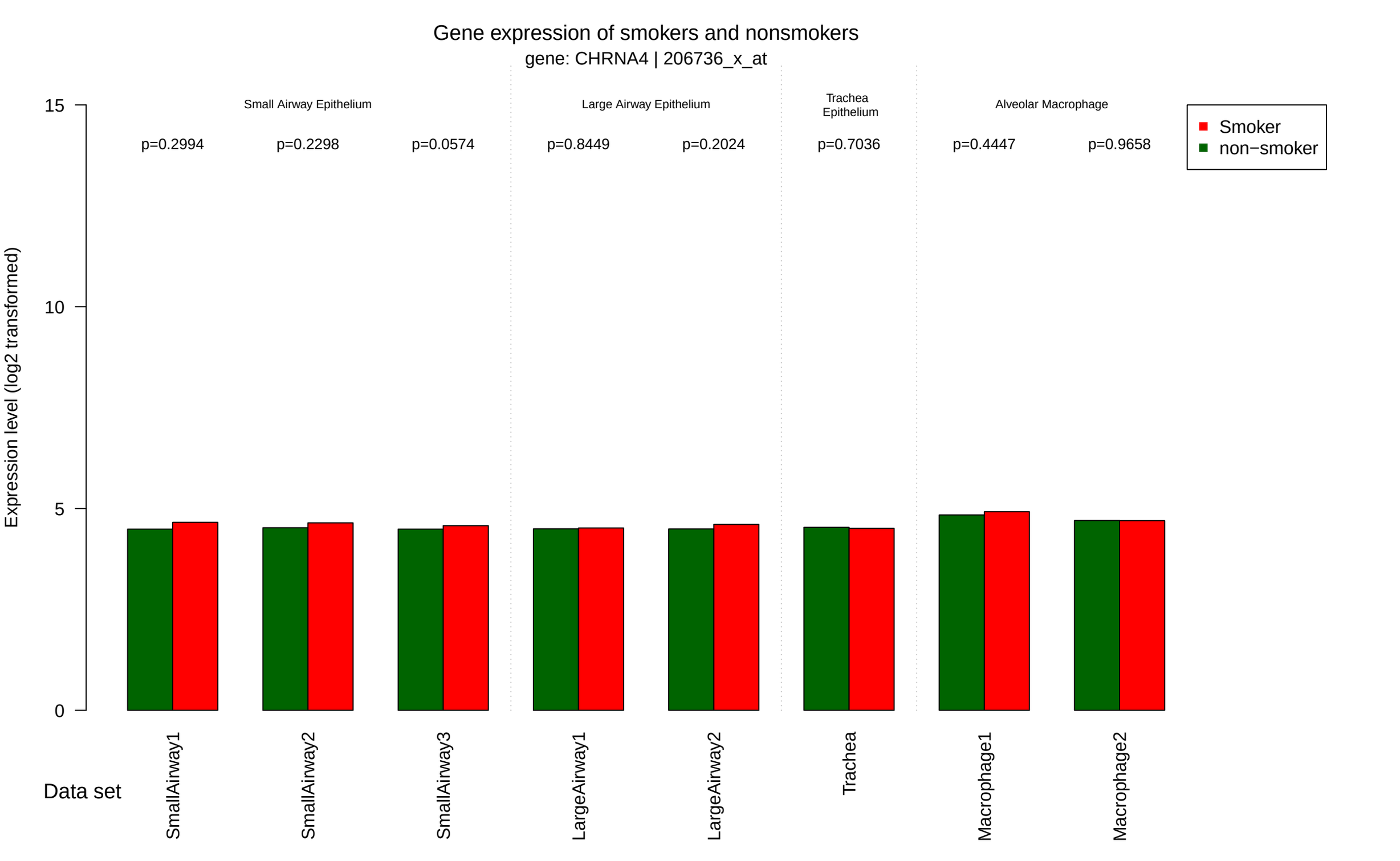
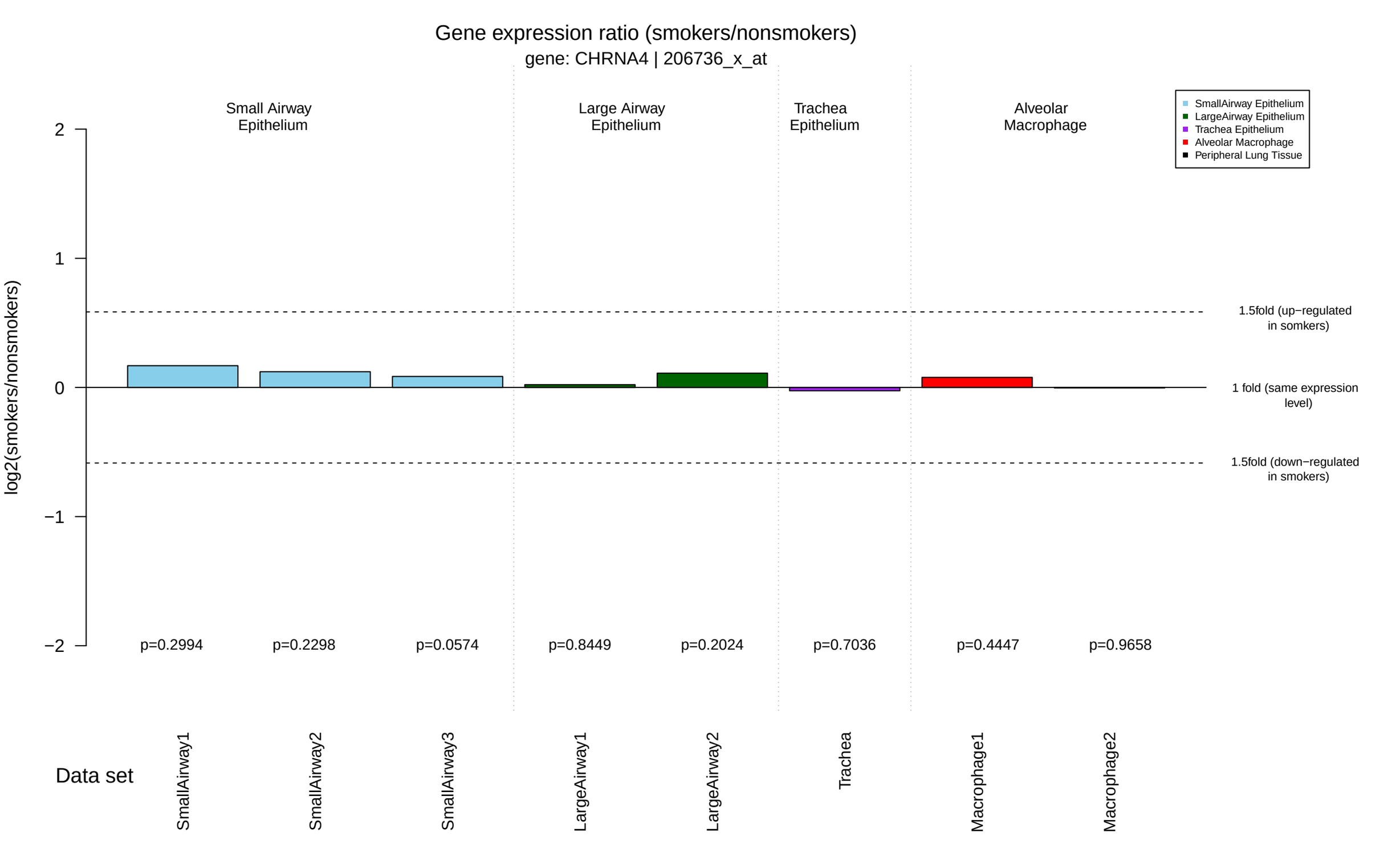
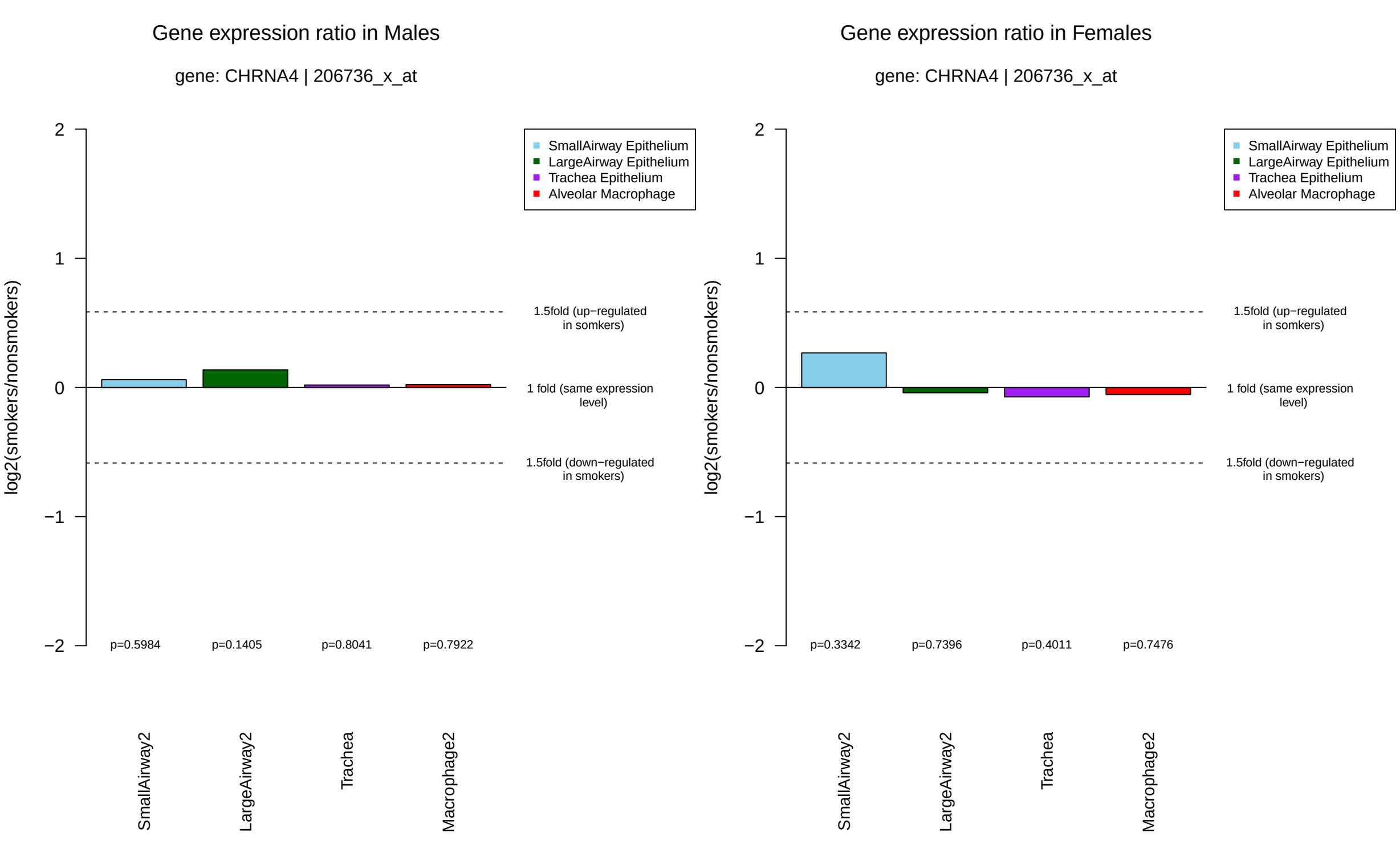
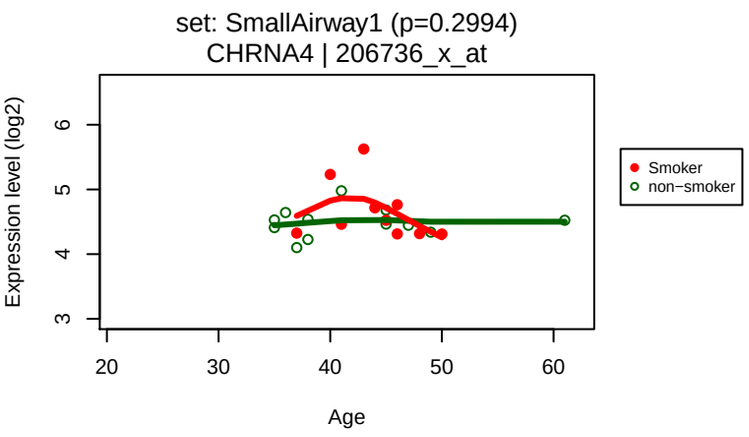
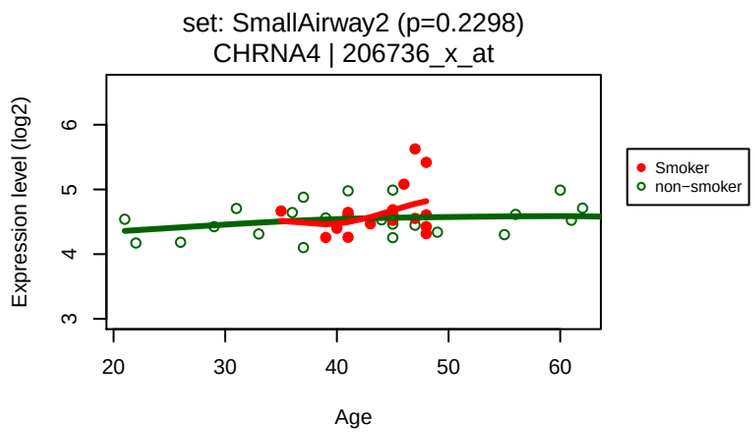
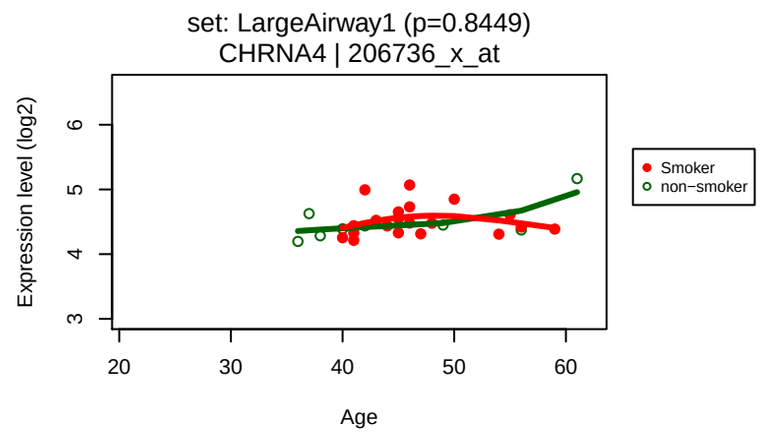

Gene expression of smokers and nonsmokers
gene: CHRNA4 | 206736_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2994
p=0.2298
p=0.0574
p=0.8449
p=0.2024
p=0.7036
p=0.4447
p=0.9658
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

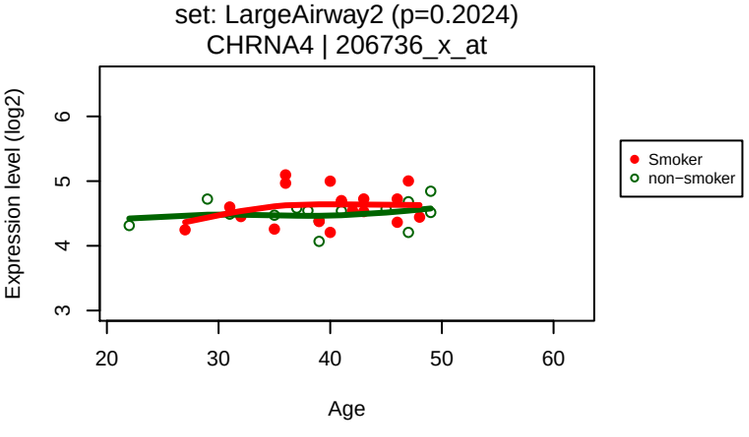

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA4 | 206736_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2994
p=0.2298
p=0.0574
p=0.8449
p=0.2024
p=0.7036
p=0.4447
p=0.9658
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

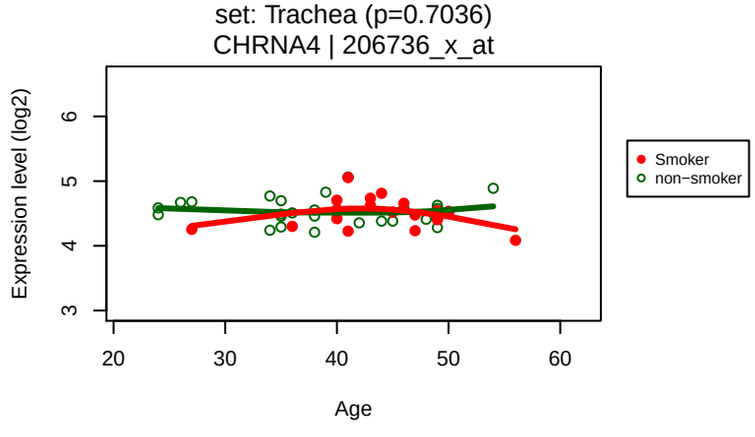

Gene expression ratio in Males
gene: CHRNA4 | 206736_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5984
p=0.1405
p=0.8041
p=0.7922
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA4 | 206736_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3342
p=0.7396
p=0.4011
p=0.7476
SmallAirway2
LargeAirway2
Trachea
Macrophage2

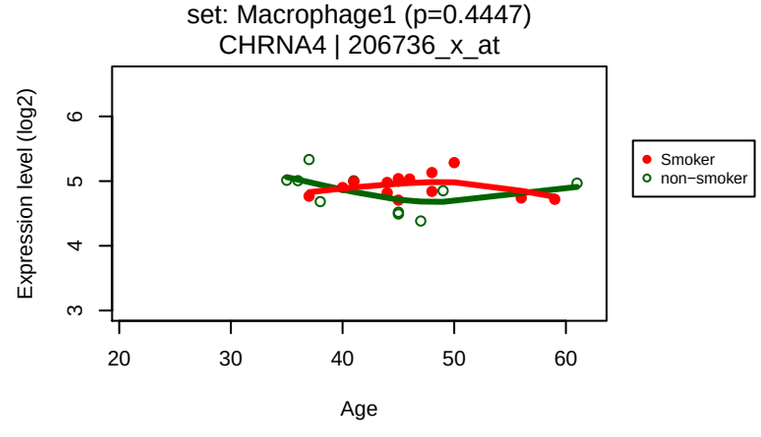

set: SmallAirway1 (p=0.2994)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker

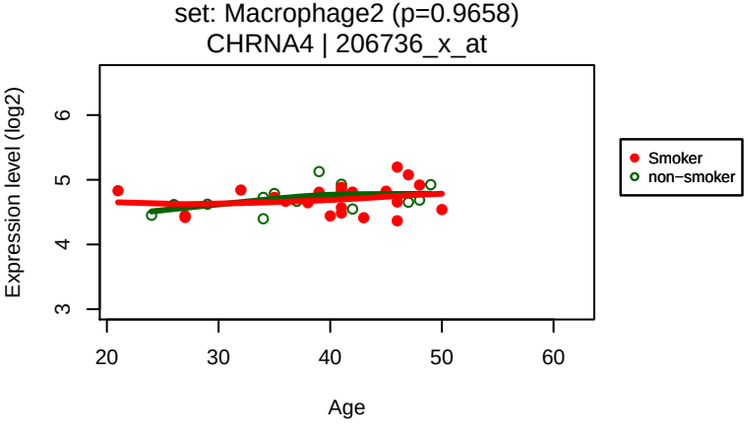

set: SmallAirway2 (p=0.2298)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker

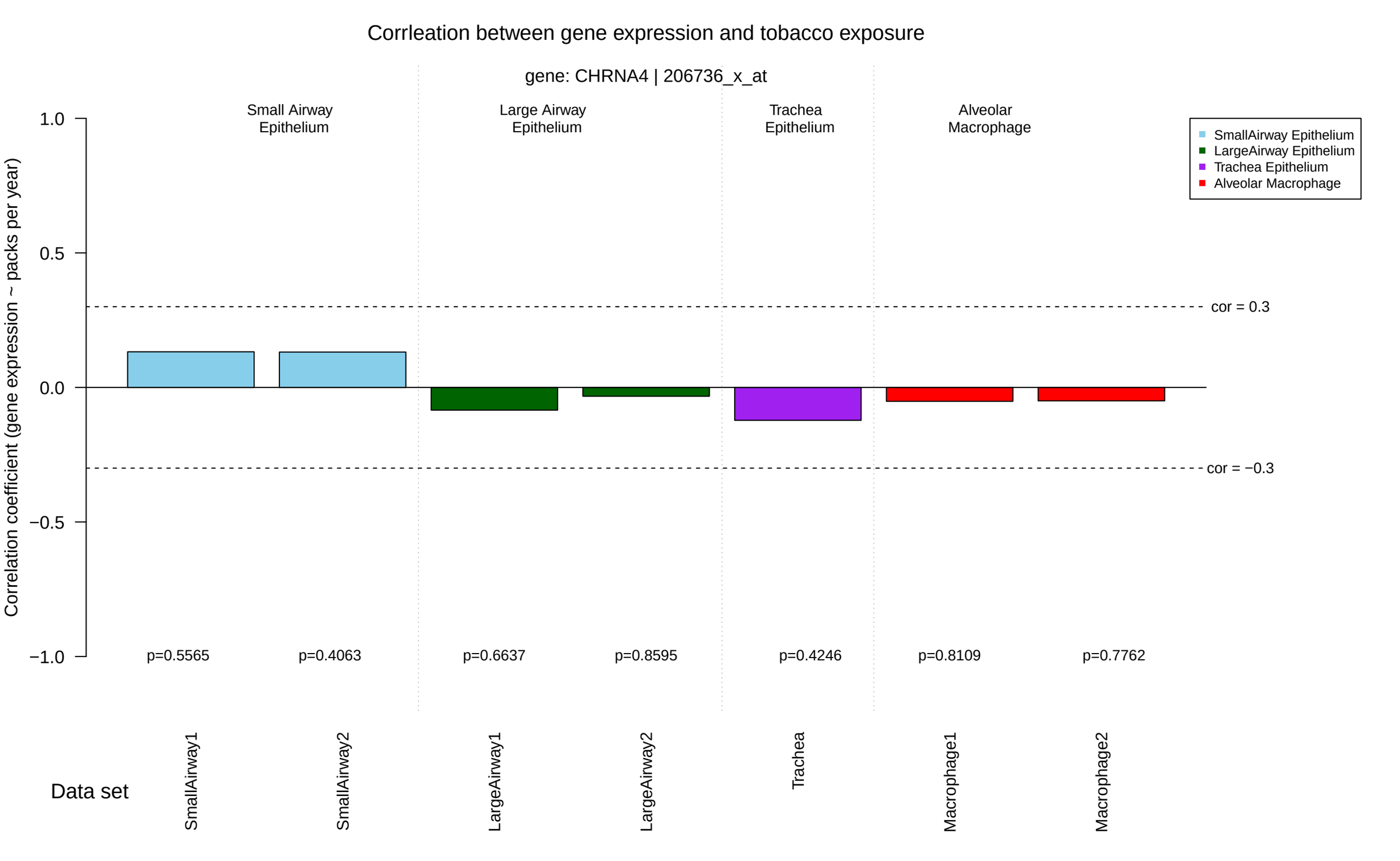

set: LargeAirway1 (p=0.8449)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2024)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7036)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4447)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9658)
CHRNA4 | 206736_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA4 | 206736_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5565
p=0.4063
p=0.6637
p=0.8595
p=0.4246
p=0.8109
p=0.7762
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2